TITLE PAGE

- **Problem Statement Title - SAFESTEP**

  (Smart Maternal Safety & Monitoring System using IoT)

- **Track - Product / Technology Solution**

- **Theme - Healthcare & Women’s Safety**

- **Team Name - INFINITY 07**





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IDEA TITLE

Problem Statement

- Helps let caregivers know when a possible fall happens.
- Reduces alerts by having a way to cancel them.
- Helps with help based on where someone's when GPS and internet are working.
- Gives a dashboard for patient records, appointments, reminders and checkups, for mothers.

EXISTING SOLUTIONS STATE THAT

37.1% of pregnant women in a southern Indian study reported foot pain during the third trimester.

Innovation and uniqueness of the solution

- Integrates wearable fall detection with a maternal-care platform.
- Combines real-time event alerts, location information and emergency history.
- Connects emergency monitoring, with appointment and follow-up support.
- Designed using IoT components for potential low-cost implementation.

Proposed Solution

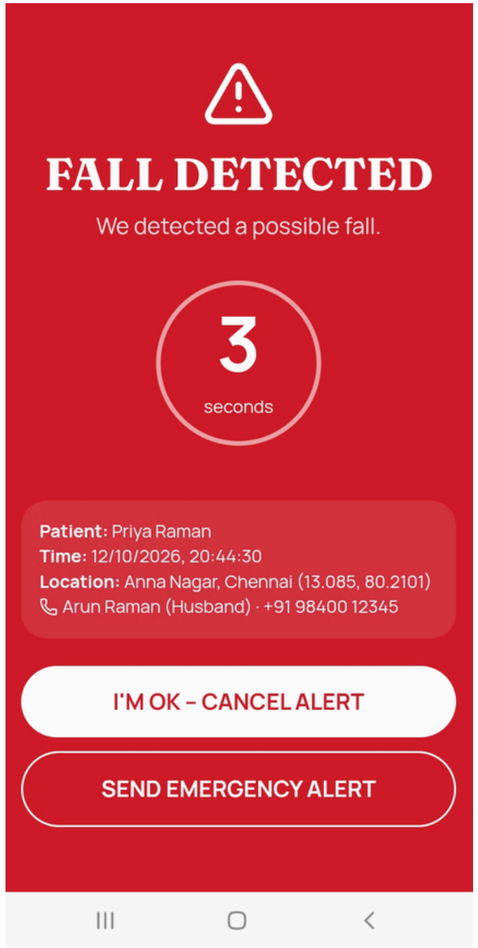
Detailed explanation of the proposed solution

- A wearable system based on IoT for detection of **emergency fall** during pregnancy.
- The system detects abnormal movement using an MPU6050 motion sensor and Seeed Studio MCU
- Triggers an instant local alert by displaying a message and vibrating.
- User can cancel false alarms before emergency notification.
- **Sends emergency alerts** with the available GPS location to registered contacts via a connected web platform..



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TECHNICAL APPROACH



Wearable Hardware

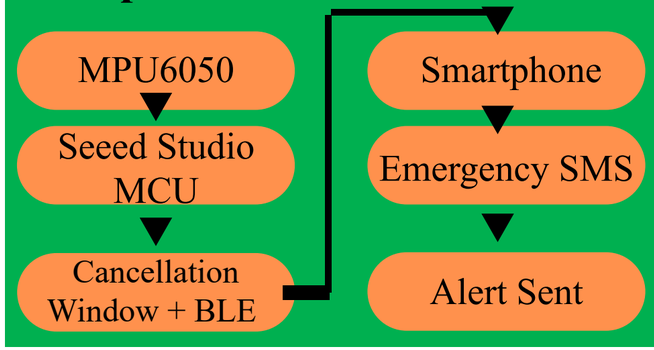
- Seeed Studio MCU: Processes sensor data and controls device responses.
- MPU6050: Measures acceleration and angular velocity for potential fall detection.
- OLED display: Displays device status and alert prompts.
- Push button: Enables alert cancellation.
- GPS module and antenna: Support location acquisition where configured.
- Rechargeable battery + charging module + 1S BMS: Support portable, rechargeable operation.
- RTC module: Support event timekeeping and wearable integration.

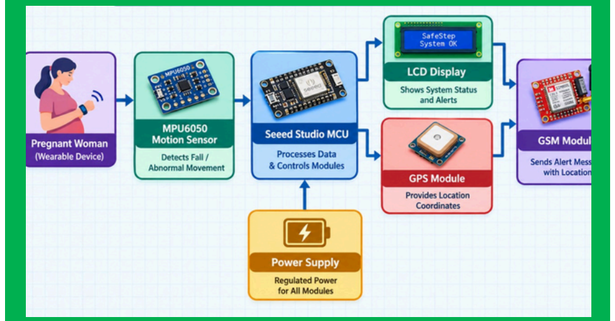


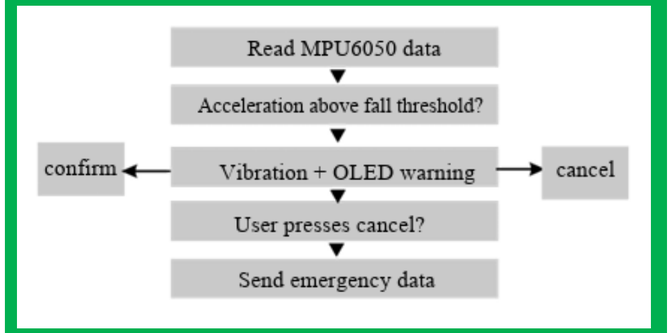
Digital Healthcare Platform

- Patient and doctor login.
- Pregnancy progress and health information.
- Diet recommendations, medication reminders and relaxation audio.
- Appointment management and doctor–patient communication.
- Speech-to-text-assisted consultation notes for doctor review.

Proposed Model







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COMPARATIVE ANALYSIS

“Falls During Pregnancy: Can cause serious injury to both mother and baby. Delayed Emergency Response: Every minute matters when help is not available quickly”.

1. Feasibility Analysis

Technical Feasibility

- Uses readily available IoT components (Seeed Studio MCU, MPU6050, GPS).
- Modular hardware and software enable easy testing.
- React and Node.js support web dashboard development.

Cost Feasibility

- Uses affordable, commercially available components.
- Suitable for prototype development and future scaling.

Operational Feasibility

- Enables local alerts and connected emergency notifications.
- Can be tested through controlled movement simulations.

2. Potential Challenges & Risks

False Alarms

- Sudden or normal movements may trigger alerts.

GPS Unavailability

- Indoor environments may reduce location accuracy.

Connectivity & Power Issues

- Network failure or battery drain may interrupt alerts.

Data Privacy & Security

- Personal and location data require protection.

User Comfort

- The wearable must be lightweight and easy to use.

3. Strategies & Mitigation

Reduce False Alarms

- Calibrate motion thresholds.
- Add a cancellation button and short alert window.

Improve Location Accuracy

- Display GPS status on the dashboard.
- Show the last known location when available.

Ensure Reliable Connectivity

- Provide local vibration alerts during network failure.
- Queue event data for transmission when connectivity returns.

Secure Patient Data

- Use authenticated access and encrypted communication.
- Restrict access to authorized users.

Enhance Usability

- Design a compact, lightweight wearable.
- Use simple alerts and easy-to-operate controls.

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Results & Discussion

Who Benefits?

- Pregnant women: Wearable fall alerts and access to maternal-care support.
- Families and guardians: Emergency SMS notifications when successfully delivered.
- Doctors: A digital channel for consultation notes and authorized patient information.
- Healthcare providers: Potential support for organized pregnancy follow-up.

Social Impact

- Helps reduce delays in communicating a possible fall-related emergency.
- Connects wearable safety with digital maternal-care services.
- Encourages continuity of care through appointments and pregnancy records.
- Supports access to organized health information and consultation documentation.

Economic Benefits

- Prototype built around modular electronic components.
- Smartphone-assisted communication avoids requiring a dedicated cellular module in the wearable for the described SMS workflow.
- Web-based architecture can be extended as requirements grow.



Who Benefits?

SAFETY : Fall-event detection and local alerts
ASSISTANCE : Smartphone-mediated emergency communication
CONTINUITY : Digital maternal-care support
SCALABILITY : Potential expansion through healthcare partnerships

Long-Term Vision

Make connected maternal safety more accessible through a wearable device and a supporting digital care platform.



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RESEARCH AND REFERENCES

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Technical References

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2. React Documentation- https://react.dev/
3. Node.js Documentation - https://nodejs.org/docs/latest/api/
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Video Link : https://drive.google.com/drive/folders/1LVgboKd_M7EOZMg4vEQw0lC9l_blPG35